**Project Report: Homework 4 Handwritten Digit Recognition using Hu Moments**


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## 1. Introduction

The objective of this assignment is to implement and analyze machine learning models for handwritten digit recognition using the MNIST dataset. Based on the applications described in *Embedded Machine Learning with Microcontrollers (Ünsalan et al., 2025)*, two distinct approaches were implemented:

1. **Binary Classification:** Using a Single Neuron model to distinguish the digit "0" from all other digits.
2. **Multi-Class Classification:** Using a Multi-Layer Perceptron (MLP) to classify all digits from "0" to "9".

Unlike standard deep learning approaches that use raw pixel data, this project utilizes **Hu Moments** as feature vectors to reduce computational complexity and achieve geometric invariance (scale, rotation, and translation).

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## 2. Data Preprocessing & Feature Extraction

The MNIST dataset was loaded using the TensorFlow/Keras API. Instead of using the raw $28 \times 28$ pixel grids, the following preprocessing steps were applied:

1. **Hu Moments Calculation:** For each image, 7 Hu Moments were calculated using the OpenCV library (cv2.HuMoments). These moments provide a compact representation of the image shape.
2. **Z-Score Normalization:** Since raw Hu Moments have vastly different magnitudes (some very small, some large), the features were normalized to have a mean of 0 and a standard deviation of 1. This step was crucial for the convergence of the Gradient Descent optimizer.

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## 3. Question 1: Single Neuron Binary Classification

**Objective:** To classify images into two categories: "Class 0" (Digit 0) and "Class 1" (Digits 1-9).

### 3.1. Methodology

- **Model Architecture:** A single dense layer with 1 neuron.
- **Activation Function:** Sigmoid (suitable for binary output).
- **Loss Function:** Binary Cross-Entropy.

- **Class Weighting:** Applied to handle class imbalance (since "Not-0" samples heavily outnumber "0" samples).

### 3.2. Results

The model was trained for 50 epochs. The performance on the test set is visualized in the Confusion Matrix below.

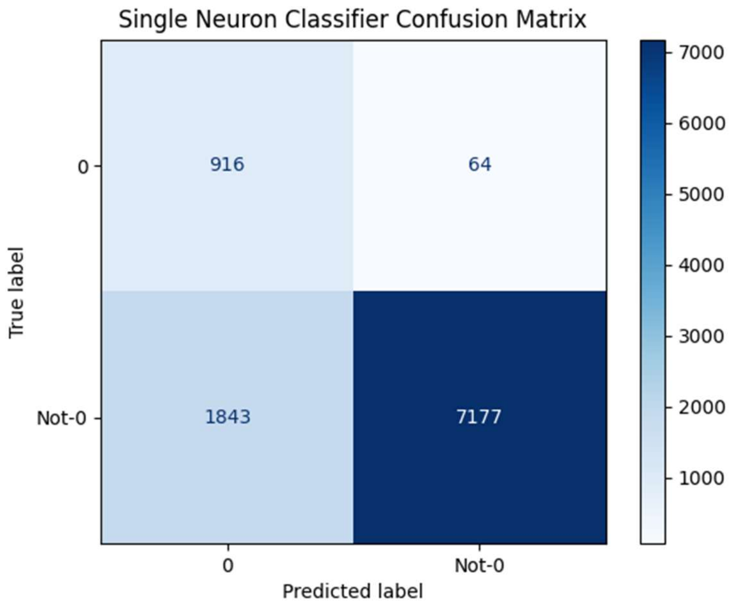


Figure 1: Confusion Matrix for Single Neuron Binary Classification.

### 3.3. Discussion

The Single Neuron achieved a high **Recall** for the target class. As seen in Figure 1, it correctly identified **916** out of ~980 zero digits, missing only **64**. This indicates that the first Hu Moment (which correlates with "roundness") is a strong indicator for the digit 0.

However, the **Precision** was low. The model incorrectly classified **1843** non-zero digits as "0". This limitation is expected because a single neuron creates a linear decision boundary. Many digits (like 6, 8, 9) share similar geometric "roundness" with 0, and a linear model using only shape moments cannot effectively separate these overlapping features.

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### 4. Question 2: Multi-Layer Neural Network (MLP)

**Objective:** To classify images into ten categories (Digits 0-9).

### 4.1. Methodology

To overcome the limitations of the single neuron, a Multi-Layer Perceptron was implemented with the following architecture:

- **Input:** 7 Normalized Hu Moments.
- **Hidden Layers:** Two layers with 100 neurons each, using **ReLU** activation.
- **Output Layer:** 10 neurons with **Softmax** activation.

- **Optimizer:** Adam (Learning rate: 1e-4).

### 4.2. Results

The model was trained with Early Stopping to prevent overfitting. The classification performance is detailed in the Confusion Matrix below.

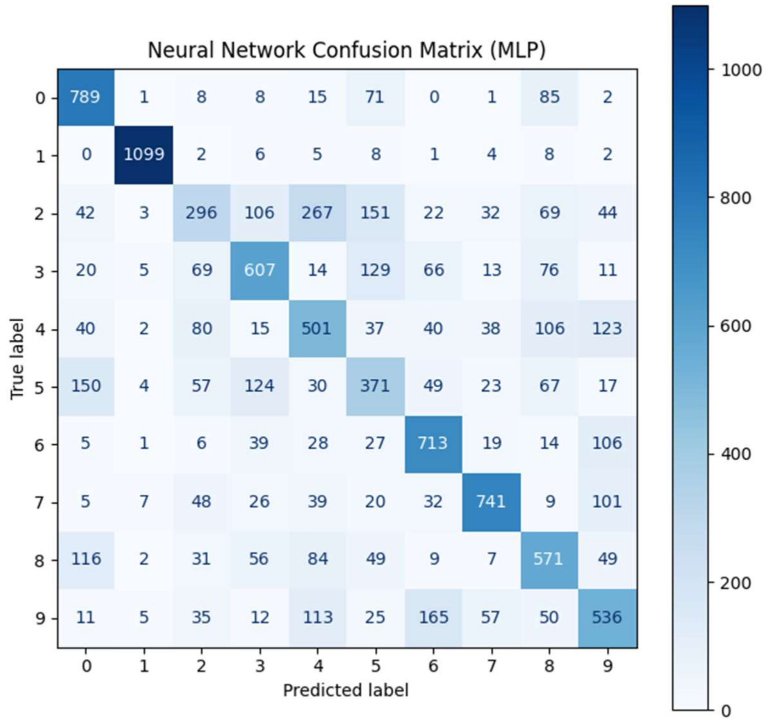


Figure 2: Confusion Matrix for MLP (10-Class Classification).

### 4.3. Discussion

The MLP provided a deeper insight into the discriminative power of Hu Moments:

- **Best Case (Digit '1'):** The model performed exceptionally well on digit '1', correctly classifying **1099** instances. The geometric shape of '1' is distinctively thin and lacks loops, making its Hu Moments unique compared to other digits.
- **Worst Case (Digit '2'):** The model struggled significantly with digit '2', often confusing it with '4' and '5'.
- **Rotation Invariance Issue:** A notable error cluster is observed between digits **'6'** and **'9'**. The model misclassified '9' as '6' **165 times**. Since Hu Moments are designed to be rotation invariant, a '6' and a rotated '9' look mathematically similar to the model, causing this confusion.

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### 5. Conclusion

This project demonstrated the trade-off between computational efficiency and classification accuracy.

1. **Efficiency:** Using Hu Moments reduced the input dimension from 784 features (pixels) to just 7. This allows the model to be extremely lightweight, suitable for the microcontrollers mentioned in the course textbook.
2. **Accuracy:** While the MLP improved upon the Single Neuron, the overall accuracy is limited by the feature extraction method. Hu Moments discard spatial details required to distinguish geometrically similar digits (like 2 vs. 5 or 6 vs. 9).

In conclusion, while Hu Moments are effective for simple shape sorting, high-precision digit recognition would require either raw pixel inputs or more complex feature descriptors.